

A Project Report On...

AgriConnect: Digital Agricultural Marketplace & Direct Farmer-to-Buyer Platform



Developed By...

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(BCA Semester - 5)**

Submitted To..

**Geetanjali College Of Computer Science And Commerce (B.C.A.)
Rajkot**

Saurashtra University Rajkot

Academic Year: 2025 - 2026

Project Guide : PROF. Kishor Vala

Acknowledgements

- I am happy to submit my idea and implementation for the "AgriConnect - Agricultural Marketplace System" web application to Saurashtra University, Rajkot for a BCA degree in the Computer Science branch.
- I am also deeply grateful to Prof. Brijesh Shah, Head of the Department, and all faculty members of the Department of Computer Science for their kind guidance, constant encouragement, and technical support throughout this journey.
- I take the privilege to acknowledge the authors of numerous reference books, research publications, Django official documentation, and technical web blogs which we referred to during the development of this project.
- I express my heartfelt gratitude to my parents and family for their unwavering moral and financial support, without which I could accomplish nothing in my academic pursuits and personal life.
- The feeling of gratefulness for everyone's timely assistance directly arises from the bottom of my heart. Small but critical guidance during testing and database schema design proved to be significant milestones in completing this application.



Preface

- Practical training and software implementation are of paramount importance in understanding theoretical aspects of Computer Application & Web Software Development. Recognizing this importance, we prepared the "AgriConnect" project to enrich our practical knowledge regarding web technologies, full-stack architecture, and database management systems.
- We are very much pleased to present this comprehensive project report on "AgriConnect: Digital Agricultural Marketplace", developed at Geetanjali College of Computer Science and Commerce, affiliated with Saurashtra University.
- This report contains an exhaustive technical and functional overview of the entire project. Anyone with basic computer software knowledge can easily understand the contents of this report. System design is illustrated using Data Flow Diagrams (DFDs), Entity Relationship (ER) diagrams, exact Data Dictionary schema definitions, test cases, and screen illustrations.
- We have tried our best in practical study, systematic code structuring, and clear presentation of this report.



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1. Project Profile

Project Title:	AgriConnect: Digital Agricultural Marketplace System
Development Software:	Visual Studio Code, Python 3.12+, Django 5.x Framework
Front End:	HTML5, CSS3, JavaScript (ES6+), Bootstrap 5, FontAwesome
Backend / Logic:	Python, Django Web Framework (MVT Architecture)
Database:	MySQL Database Management System
Academic Year:	2025 - 2026
Developed By:	Smit Pithadiya (BCA Semester - 6)
Submitted To:	Geetanjali College Of Computer Science And Commerce, Rajkot
Documentation Tool:	Microsoft Word, Draw.io, Python Docx, Matplotlib
Operating System:	Cross-platform (Windows 10/11, macOS, Linux)
Programming Language:	Python 3.x, JavaScript, HTML, CSS

2. Software Development Life Cycle (SDLC)

The Software Development Life Cycle (SDLC) is a systematic, structured methodology utilized to design, develop, test, and deploy high-quality web software applications. For the development of the AgriConnect project, the Waterfall Model was selected due to its linear, sequential flow and well-defined phase dependencies.

Waterfall Model Steps:

- 1. Requirement Gathering & Feasibility Analysis
- 2. Project Planning & Scope Definition
- 3. System Design & Architecture Specification
- 4. Coding & Full-Stack Implementation
- 5. Testing & Quality Assurance

Phase 1: Requirement Gathering

In this initial stage, detailed functional and non-functional requirements were gathered by researching existing agricultural trade platforms and consulting local farmers and produce buyers.

- • Farmer Requirements: Registration, identity verification, crop listing management, quantity tracking, price per kg setup, stock status, direct buyer order reception, earnings summary, and direct chat.
- • Buyer Requirements: Marketplace browsing, category filtering, search, crop details, shopping cart management, delivery address entry, multi-mode payment, order placement with OTP verification, order history, and farmer reviews.
- • Admin Requirements: Centralized dashboard, crop approval workflow, user account role management, platform analytics, system reports generation, and dispute resolution.

Phase 2: Project Planning

A comprehensive project plan was established outlining timelines, milestone deadlines, module division, technology selection (Python + Django MVT), and database choice.

Phase 3: System Design

The architecture was designed using Django's Model-View-Template (MVT) pattern. Database schemas were constructed for 21 core entities including User, Farmer, Buyer, AuthOTP, Category, Crop, CartItem, Order, OrderItem, Payment, Review, Notification, ChatMessage, Wishlist, FarmerRating, Feedback, PriceTrend, MarketInsight, ReturnRequest, Report, and OrderOTP tables. Data Flow Diagrams (DFDs) and ER diagrams were modeled to guarantee clean relationships.

Phase 4: Coding & Implementation

Development was executed using Python 3 and Django 5. Web templates were built using HTML5, CSS3, JavaScript, and Bootstrap 5 to deliver a responsive user interface. Clean coding standards (PEP 8) and secure design principles (CSRF tokens, password hashing, RBAC) were strictly enforced.

Phase 5: Testing

Rigorous Unit Testing, Integration Testing, System Testing, and User Acceptance Testing (UAT) were conducted across all farmer, buyer, and admin workflows to verify functional correctness and zero defects.



3. System Requirements

To successfully deploy, run, and maintain the AgriConnect project, the following hardware and software specifications are required:

Minimum Hardware Requirements:

Component / Resource	Minimum Specification
Operating System	Windows 10/11, macOS 11+, Linux Ubuntu 20.04+
CPU / Processor	Intel Core i3 / AMD Ryzen 3 Dual-Core (2.0 GHz or higher)
System RAM	4 GB (8 GB recommended for optimal development)
Hard Disk Space	20 GB available storage space
Display Resolution	1280 x 720 HD Resolution or higher
Network Connection	Broadband Internet (for static CDN resources & deployment)

Software Requirements:

Software Tool / Component	Version / Specification
Runtime Environment	Python 3.12+ 64-bit Interpreter
Web Framework	Django 5.x Framework
Database Server	MySQL Database Server (v8.0+ / TiDB Cloud)
Frontend Stack	HTML5, CSS3, JavaScript (ES6+), Bootstrap 5.3
Web Server	Django WSGI / ASGI Dev Server (Apache 2.4 / Nginx in Production)
IDE / Source Editor	Visual Studio Code / PyCharm
Version Control	Git & GitHub Repository

4. Development & Coding Standards

To ensure maximum readability, maintainability, and scalability across the AgriConnect codebase, strict Python (PEP 8) and web engineering standards were enforced during implementation.

Naming Conventions:

- **File & Function Naming:** Use snake_case for filenames, module names, Python variables, and function names (e.g., crop_detail_view, price_per_kg).
- **Class Naming:** Use PascalCase for Django models, views, forms, and Python classes (e.g., Crop, CartItem, Wishlist).
- **Template Naming:** Use lowercase snake_case for HTML template files placed under clear app folders (e.g., templates/marketplace/crop_detail.html).

Code Formatting & Structure:

- **Indentation:** Consistently use 4 spaces per indentation level. Avoid tab characters to maintain environment consistency.
- **Line Length:** Limit Python source code lines to 79-88 characters for enhanced visual readability and side-by-side editing.
- **Import Organization:** Group imports logically: Standard library imports first, Third-party imports second, and Local app imports third.

Error Handling & Security Standards:

- **Exception Handling:** Wrap database queries, file operations, and external API requests inside try-except blocks to gracefully capture runtime exceptions.
- **Access Control:** Enforce @login_required and role-verification decorators on views to restrict unauthorized URL access.
- **Security Tokens:** Include {% csrf_token %} on all HTML POST forms to prevent Cross-Site Request Forgery attacks.

5. About the Tool (Tools & Technologies Used)

Python Programming Language:

Python is a high-level, interpreted, object-oriented programming language known for its simple syntax, readability, and extensive standard library. Python powers the backend business logic and data manipulation of AgriConnect.

Django Web Framework:

Django is a high-level Python web framework that encourages rapid development and clean, pragmatic design. Built by experienced developers, it takes care of much of the hassle of web development. Django follows the Model-View-Template (MVT) architectural pattern and provides built-in Object-Relational Mapping (ORM), robust user authentication, form validation, and admin portal capabilities out of the box.

HTML5 & CSS3 & Bootstrap 5:

HTML5 provides semantic structure for web pages, while CSS3 handles responsive visual styling. Bootstrap 5, a popular front-end CSS framework, is integrated to ensure AgriConnect renders flawlessly on desktop displays, tablets, and smartphones.

MySQL Database Management System (DBMS):

MySQL is a high-performance, robust relational database management system (RDBMS) used to store and manage all relational data in AgriConnect. Structured database tables store user credentials, farmer profiles, crop inventories, shopping cart items, order transactions, payments, reviews, and market insights. In production, MySQL / TiDB Cloud is utilized to deliver enterprise-grade performance, ACID transaction safety, foreign key integrity, and high-concurrency reliability.

Visual Studio Code (VS Code):

VS Code is a lightweight but powerful source code editor running on desktop OS. It features built-in support for Python, HTML/CSS, Git version control, syntax highlighting, and interactive debugging.

6. Data Flow Diagrams (DFD)

Data Flow Diagrams graphically illustrate how data flows through the AgriConnect system, identifying external entities, process transformations, and internal database stores.

Data Flow Diagram (DFD Level 0 - Context Diagram):

Data Flow Diagram (DFD Level 0) - AgriConnect Context Diagram

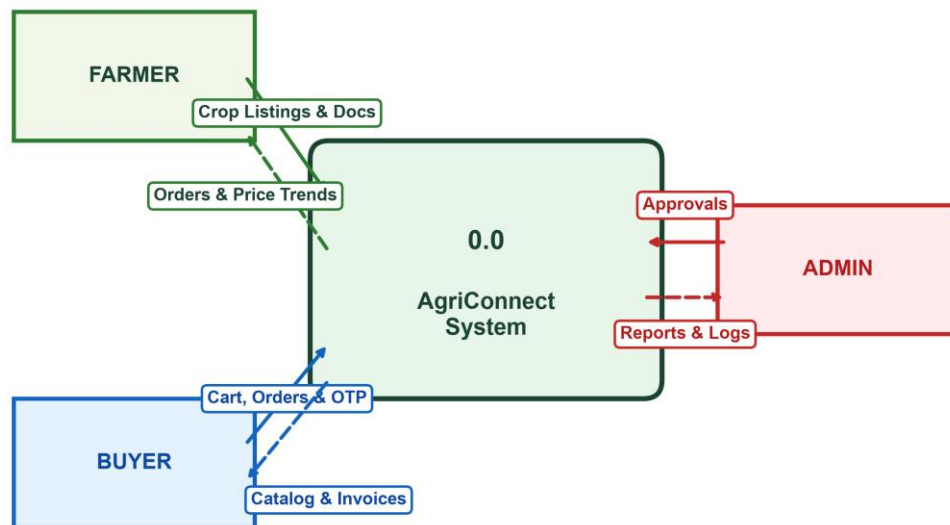


Figure 6.1: Context Diagram (DFD Level 0) illustrating high-level interactions between Farmers, Buyers, Admins, and the central AgriConnect platform.

Data Flow Diagram (DFD Level 1 - Process Breakdown):

Data Flow Diagram (DFD Level 1) - AgriConnect Subsystems Breakdown

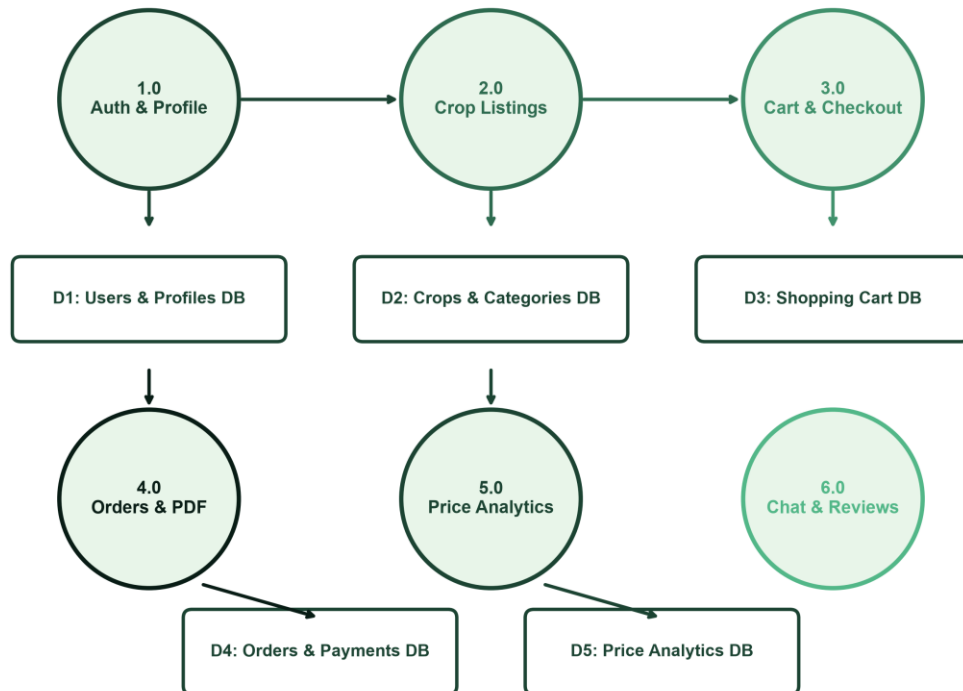


Figure 6.2: Detailed Process Breakdown (DFD Level 1) showing data paths between authentication, crop listing, shopping cart, order placement with OTP, payment, and database stores.

7. Entity Relationship (ER) Diagram & Architecture

Entity Relationship (ER) Diagram:

AgriConnect Entity Relationship (ER) Diagram - Core Relational Schema

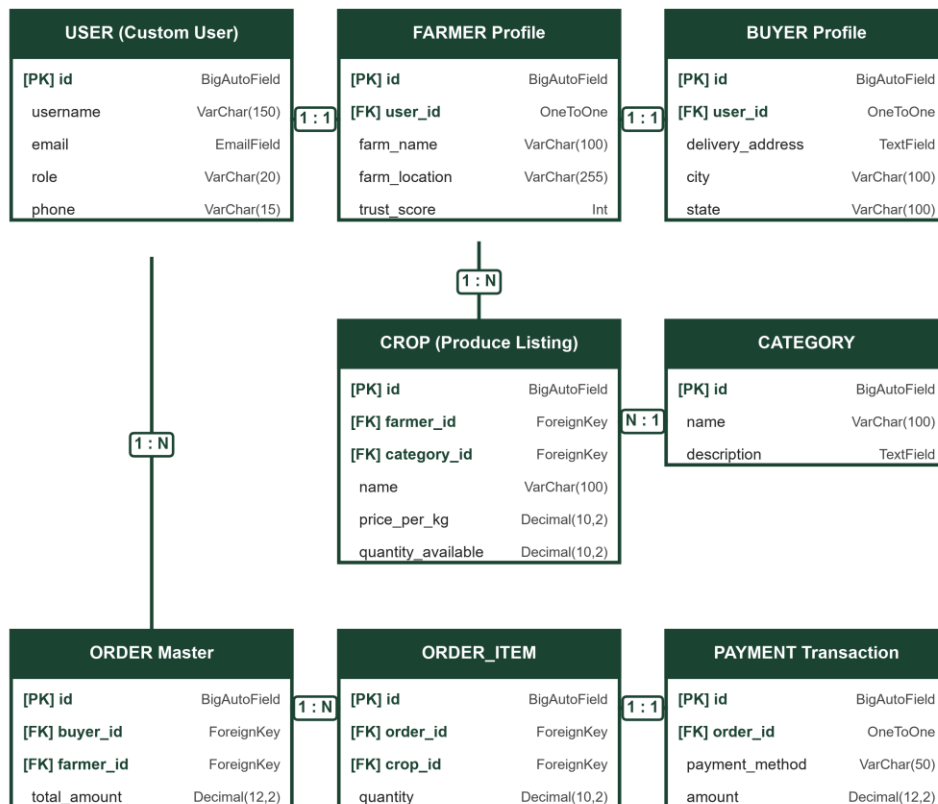


Figure 7.1: Entity Relationship Diagram depicting primary keys, foreign key constraints, and cardinalities between User, Crop, Category, Order, OrderItem, and Payment entities.

System Architecture Diagram:

System Architecture Diagram - Django MVT Pattern

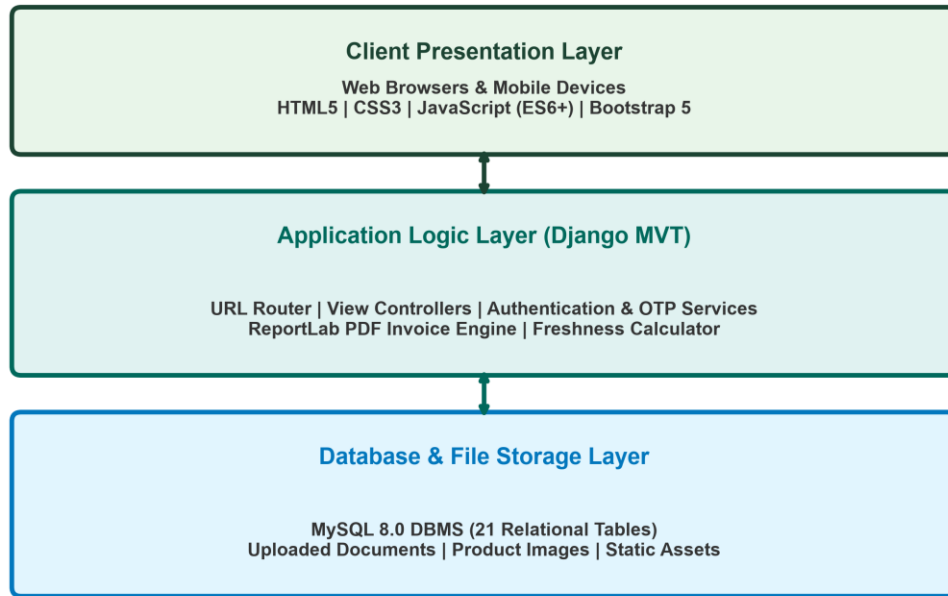


Figure 7.2: 3-Tier Architecture showing Presentation Layer, Django Application Server Layer, and Relational Database Storage Layer.

Use Case Diagram:

Use Case Diagram - AgriConnect System Roles

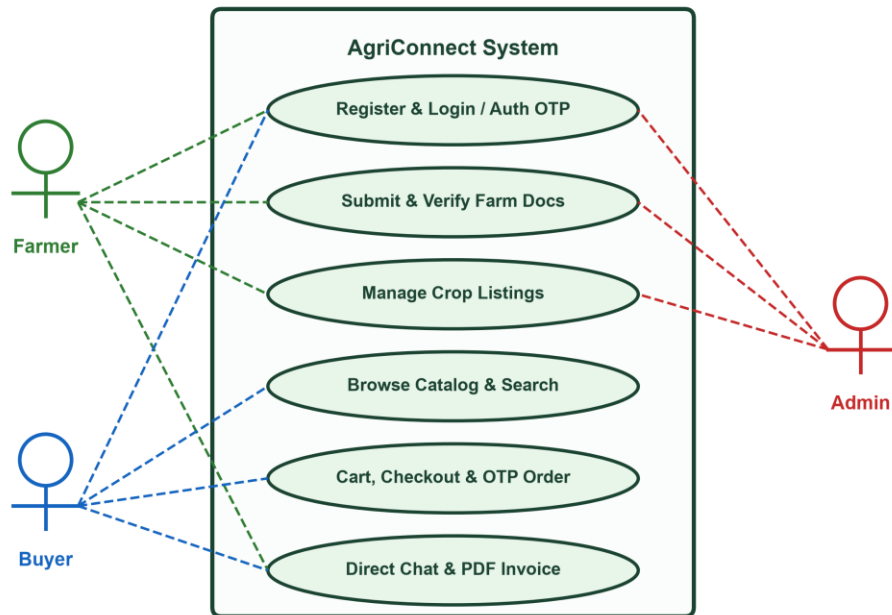


Figure 7.3: Use Case Diagram depicting interactions between Farmer, Buyer, and Admin roles.

8. Data Dictionary

The Data Dictionary provides the exact schema definition for all 21 database tables implemented in the AgriConnect relational database system, detailing field names, data types, nullability, key constraints, and functional descriptions.

Table 1: accounts_user (Custom User Model)

Field Name	Data Type	Null?	Constraint	Description
id	BigAutoField	No	Primary Key	Unique user identifier
username	CharField	No	Unique	Login username
email	EmailField	No	Unique	User email address
role	CharField	No	-	User role (admin, farmer, buyer)
phone	CharField	Yes	-	Contact phone number
address	TextField	Yes	-	Physical delivery/farm address
profile_picture	ImageField	Yes	-	Profile avatar image
is_active	BooleanField	No	-	Account active status flag
date_joined	DateTimeField	No	-	Account registration timestamp

Table 2: accounts_farmer (Farmer Profile)

Field Name	Data Type	Null?	Constraint	Description
id	BigAutoField	No	Primary Key	Profile ID
user_id	OneToOneField	No	FK (User)	Reference to

				accounts_user
farm_name	CharField	Yes	-	Name of the farm / enterprise
farm_location	CharField	Yes	-	Farm address / region
farm_size	CharField	Yes	-	Land acreage (e.g. 5 Acres)
certification_status	CharField	No	-	Verification badge status (verified/pending/none)
experience	CharField	No	-	Years of agricultural experience
orders_completed	IntegerField	No	-	Count of fulfilled orders
store_rating	FloatField	No	-	Average store rating (0.0 - 5.0)
specialization	CharField	No	-	Farming domain specialization
about	TextField	Yes	-	Farmer biography & overview
cover_banner	ImageField	Yes	-	Storefront header cover banner
certifications	TextField	Yes	-	Agricultural certifications held
awards	TextField	Yes	-	Recognitions and awards received
repeat_customers	IntegerField	No	-	Count/rate of repeat buyers
ontime_delivery	IntegerField	No	-	On-time delivery percentage
satisfaction_rate	IntegerField	No	-	Customer satisfaction percentage
village	CharField	Yes	-	Village name

city	CharField	Yes	-	City location
state	CharField	Yes	-	State location
pincode	CharField	Yes	-	Postal code
organic_certificate	FileField	Yes	-	Uploaded organic farming certificate
verification_status	CharField	No	-	Admin verification status choice
aadhaar_document	FileField	Yes	-	Identity verification document
farm_certificate	FileField	Yes	-	Land certificate document
trust_score	IntegerField	No	-	Platform trust rating score
verification_date	DateTimeField	Yes	-	Verification approval timestamp
verified_by_id	ForeignKey	Yes	FK (User)	Admin user performing verification
admin_notes	TextField	Yes	-	Verification review notes
approval_history	TextField	Yes	-	Audit trail log of actions

Table 3: accounts_buyer (Buyer Profile)

Field Name	Data Type	Null?	Constraint	Description
id	BigAutoField	No	Primary Key	Profile ID
user_id	OneToOneField	No	FK (User)	Reference to accounts_user
delivery_address	TextField	Yes	-	Primary shipping destination
contact_name	CharField	Yes	-	Contact person name

city	CharField	Yes	-	Buyer city location
state	CharField	Yes	-	Buyer state location
pincode	CharField	Yes	-	Postal pincode

Table 4: accounts_authotp (Authentication OTP)

Field Name	Data Type	Null?	Constraint	Description
id	BigAutoField	No	Primary Key	OTP Record ID
user_id	ForeignKey	Yes	FK (User)	Target user reference
email	EmailField	Yes	-	Email address for unregistered OTP
otp_code	CharField	No	-	6-digit OTP passcode
otp_type	CharField	No	-	OTP type (registration / password_reset)
created_at	DateTimeField	No	-	OTP generation timestamp
expires_at	DateTimeField	No	-	OTP expiration timestamp
is_verified	BooleanField	No	-	OTP verification status flag
attempts	IntegerField	No	-	Failed attempt counter
resend_count	IntegerField	No	-	Resend request counter

Table 5: marketplace_category (Crop Category)

Field Name	Data Type	Null?	Constraint	Description
id	BigAutoField	No	Primary Key	Category ID
name	CharField	No	Unique	Category title (Vegetables, Fruits, Grains)
description	TextField	Yes	-	Category detailed description
image	ImageField	Yes	-	Category banner thumbnail image

Table 6: marketplace_crop (Crop Product Listing)

Field Name	Data Type	Null ?	Constraint	Description
id	BigAutoField	No	Primary Key	Crop Product ID
farmer_id	ForeignKey	No	FK (Farmer)	Farmer owner of crop listing
category_id	ForeignKey	No	FK (Category)	Produce category reference
name	CharField	No	-	Crop name (e.g. Organic Wheat)
description	TextField	No	-	Detailed produce description
price_per_kg	DecimalField	No	-	Price per unit / kilogram (INR)
quantity_available	DecimalField	No	-	Available stock quantity
unit	CharField	No	-	Unit choice (kg, ton, piece, quintal)
image	ImageField	Yes	-	Produce primary photograph
is_approved	BooleanField	No	-	Admin approval status

harvest_date	DateField	No	-	Date crop was harvested
shelf_life_days	IntegerField	No	-	Estimated shelf life in days
availability_status	CharField	No	-	Stock status (available/out_of_stock)
created_at	DateTimeField	No	-	Listing creation timestamp
updated_at	DateTimeField	No	-	Listing last update timestamp
status_history	JSONField	No	-	Audit trail of listing changes
otp_code	CharField	Yes	-	Verification OTP code
otp_expires_at	DateTimeField	Yes	-	OTP expiry timestamp
is_otp_verified	BooleanField	No	-	OTP verification flag
return_reason	CharField	Yes	-	Return reason choice if returned
return_description	TextField	Yes	-	Return details note
return_status	CharField	Yes	-	Return status (Pending/Approved/Rejected)
market_demand_level	CharField	No	-	Market demand index (High/Medium/Low)
market_forecast_direction	CharField	No	-	Price forecast direction
current_market_price	DecimalField	Yes	-	APMC mandi benchmark price
market_price_history	JSONField	No	-	Price history trend data

Table 7: marketplace_cartitem (Shopping Cart Item)

Field Name	Data Type	Null?	Constraint	Description
id	BigAutoField	No	Primary Key	Cart Item ID

user_id	ForeignKey	No	FK (User)	Buyer user owning cart
crop_id	ForeignKey	No	FK (Crop)	Selected crop item
quantity	PositiveInt	No	-	Selected quantity count

Table 8: marketplace_order (Order Master Table)

Field Name	Data Type	Null?	Constraint	Description
id	BigAutoField	No	Primary Key	Order ID
buyer_id	ForeignKey	No	FK (User)	Buyer placing the order
farmer_id	ForeignKey	No	FK (Farmer)	Farmer fulfilling order
total_amount	DecimalField	No	-	Total payable order cost (INR)
payment_method	CharField	No	-	Payment method choice (COD/Online)
status	CharField	No	-	Order status (Pending/Confirmed/Delivered)
shipping_address	TextField	No	-	Complete shipping destination
invoice_pdf	FileField	Yes	-	Generated PDF tax invoice
created_at	DateTimeField	No	-	Order placement timestamp
updated_at	DateTimeField	No	-	Order last modified timestamp
status_history	JSONField	No	-	Order state lifecycle history
otp_code	CharField	Yes	-	6-digit delivery confirmation OTP
otp_expires_at	DateTimeField	Yes	-	OTP expiry timestamp
is_otp_verified	BooleanField	No	-	OTP verified status flag
return_reason	CharField	Yes	-	Order return reason
return_description	TextField	Yes	-	Order return details

return_status	CharField	Yes	-	Return request approval status
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Table 9: marketplace_orderitem (Order Line Items)

Field Name	Data Type	Null?	Constraint	Description
id	BigAutoField	No	Primary Key	Line Item ID
order_id	ForeignKey	No	FK (Order)	Parent order reference
crop_id	ForeignKey	Yes	FK (Crop)	Purchased crop product
quantity	DecimalField	No	-	Purchased quantity volume
price_per_unit	DecimalField	No	-	Unit price at purchase time

Table 10: marketplace_payment (Payment Transaction)

Field Name	Data Type	Null?	Constraint	Description
id	BigAutoField	No	Primary Key	Payment ID
order_id	OneToOneField	No	FK (Order)	Associated order reference
payment_method	CharField	No	-	Payment mode (COD / Online / UPI)
transaction_id	CharField	Yes	-	Gateway transaction reference
status	CharField	No	-	Payment status (Pending/Completed/Failed)
amount	DecimalField	No	-	Transaction total amount
created_at	DateTimeField	No	-	Payment processing timestamp

Table 11: marketplace_review (Customer Crop & Farmer Review)

Field Name	Data Type	Null?	Constraint	Description
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id	BigAutoField	No	Primary Key	Review ID
buyer_id	ForeignKey	No	FK (User)	Reviewer buyer user
crop_id	ForeignKey	Yes	FK (Crop)	Reviewed crop product
farmer_id	ForeignKey	Yes	FK (Farmer)	Reviewed farmer profile
quality_rating	PositiveInt	Yes	-	Crop quality rating (1-5)
communication_rating	PositiveInt	Yes	-	Farmer response score (1-5)
delivery_rating	PositiveInt	Yes	-	Delivery speed score (1-5)
packaging_rating	PositiveInt	Yes	-	Packaging quality score (1-5)
rating	PositiveInt	No	-	Overall star rating score (1-5)
title	CharField	Yes	-	Review headline title
comment	TextField	Yes	-	Detailed feedback comment
image	ImageField	Yes	-	Customer product photo upload
reply	TextField	Yes	-	Seller response reply text
reply_at	DateTimeField	Yes	-	Seller reply timestamp
created_at	DateTimeField	No	-	Review

		submission timestamp
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Table 12: marketplace_notification (System Notification)

Field Name	Data Type	Null ?	Constraint	Description
id	BigAutoField	No	Primary Key	Notification ID
user_id	ForeignKey	No	FK (User)	Recipient user account
message	TextField	No	-	Notification text body
notification_type	CharField	No	-	Type (order/price/weather/system/chat/review/verification)
priority	CharField	No	-	Priority level (high/medium/low)
link	CharField	Yes	-	Action link URL
is_read	BooleanField	No	-	Read/unread status flag
created_at	DateTimeField	No	-	Dispatch timestamp

Table 13: marketplace_chatmessage (Buyer-Farmer Chat Message)

Field Name	Data Type	Null?	Constraint	Description
id	BigAutoField	No	Primary Key	Message ID
sender_id	ForeignKey	No	FK (User)	Sender user reference
receiver_id	ForeignKey	No	FK (User)	Recipient user reference
crop_id	ForeignKey	Yes	FK (Crop)	Associated crop produce inquiry

message	TextField	No	-	Message text content
is_read	BooleanField	No	-	Read status flag
created_at	DateTimeField	No	-	Message sent timestamp

Table 14: marketplace_wishlist (Saved Crops Wishlist)

Field Name	Data Type	Null?	Constraint	Description
id	BigAutoField	No	Primary Key	Wishlist Item ID
user_id	ForeignKey	No	FK (User)	Buyer user reference
crop_id	ForeignKey	No	FK (Crop)	Saved crop listing reference
added_at	DateTimeField	No	-	Added timestamp

Table 15: marketplace_farmerrating (Farmer Rating & Evaluation)

Field Name	Data Type	Null?	Constraint	Description
id	BigAutoField	No	Primary Key	Rating Record ID
buyer_id	ForeignKey	No	FK (User)	Rating buyer reference
farmer_id	ForeignKey	No	FK (Farmer)	Rated farmer reference
rating	PositiveInt	No	-	Overall rating score (1-5)
quality_rating	PositiveInt	No	-	Quality score (1-5)
communication_rating	PositiveInt	No	-	Communication score (1-5)
delivery_rating	PositiveInt	No	-	Delivery score

				(1-5)
packaging_rating	PositiveInt	No	-	Packaging score (1-5)
comment	TextField	Yes	-	Detailed evaluation feedback
created_at	DateTimeField	No	-	Rating timestamp

Table 16: marketplace_feedback (Platform Feedback)

Field Name	Data Type	Null?	Constraint	Description
id	BigAutoField	No	Primary Key	Feedback ID
user_id	ForeignKey	No	FK (User)	Submitting user reference
subject	CharField	No	-	Feedback subject line
message	TextField	No	-	Detailed feedback message
is_resolved	BooleanField	No	-	Admin resolution status flag
created_at	DateTimeField	No	-	Submission timestamp

Table 17: marketplace_pricetrend (Crop Market Price Trend)

Field Name	Data Type	Null?	Constraint	Description
id	BigAutoField	No	Primary Key	Trend Record ID
crop_name	CharField	No	-	Crop commodity name
category_id	ForeignKey	No	FK (Category)	Category reference

avg_price	DecimalField	No	-	Average market price per kg (INR)
record_date	DateField	No	-	Recording date

Table 18: marketplace_marketinsight (Market Forecast & Analytics)

Field Name	Data Type	Null?	Constraint	Description
id	BigAutoField	No	Primary Key	Insight Record ID
crop_name	CharField	No	-	Commodity crop name
category_id	ForeignKey	No	FK (Category)	Category reference
current_price	DecimalField	No	-	Current APMC mandi price (INR)
price_change_percent	DecimalField	No	-	Price percentage change (+/- %)
demand_level	CharField	No	-	Demand index (High/Medium/Low)
forecast_direction	CharField	No	-	Forecast trend direction
region	CharField	No	-	APMC Mandi / Region name
volume_tonnes	IntegerField	No	-	Traded volume in tonnes
record_date	DateField	No	-	Generation record date

Table 19: marketplace_returnrequest (Order Return & Refund Request)

Field Name	Data Type	Null?	Constraint	Description
id	BigAutoField	No	Primary Key	Return Request ID
order_id	ForeignKey	No	FK (Order)	Parent order reference

reason	CharField	No	-	Return reason choice
description	TextField	Yes	-	Detailed return reason description
status	CharField	No	-	Status (Pending/Approved/Rejected)
created_at	DateTimeField	No	-	Request filing timestamp

Table 20: marketplace_report (System & Sales Reports)

Field Name	Data Type	Null?	Constraint	Description
id	BigAutoField	No	Primary Key	Report Record ID
name	CharField	No	-	Report document title
report_type	CharField	No	-	Report classification type
generated_by_id	ForeignKey	No	FK (User)	Generating user reference
format	CharField	No	-	File format (pdf, csv, xlsx)
download_count	PositiveInt	No	-	Download counter
file	FileField	Yes	-	Generated file path
scheduled_interval	CharField	Yes	-	Scheduled frequency (daily, weekly, monthly)
created_at	DateTimeField	No	-	Report generation timestamp

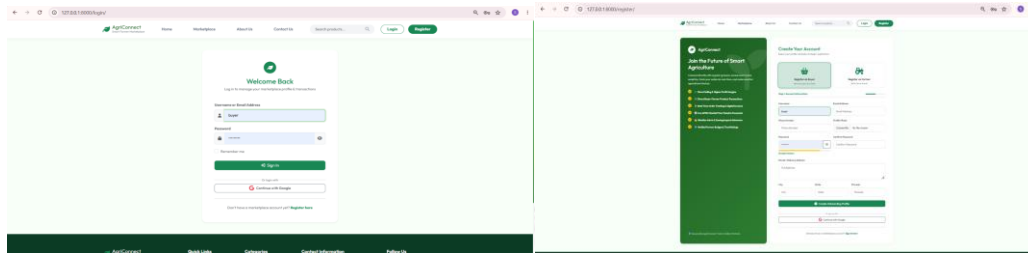
Table 21: marketplace_orderotp (Order Confirmation OTP)

Field Name	Data Type	Null?	Constraint	Description
id	BigAutoField	No	Primary Key	Order OTP Record ID
user_id	ForeignKey	No	FK (User)	Buyer user reference
otp_code	CharField	No	-	6-digit OTP passcode
shipping_address	TextField	No	-	Shipping destination address
payment_method	CharField	No	-	Selected payment method
created_at	DateTimeField	No	-	OTP generation timestamp
expires_at	DateTimeField	No	-	OTP expiration timestamp
resend_count	PositiveInt	No	-	Resend attempt counter
is_verified	BooleanField	No	-	Verification flag status

9. Screen Shots & Module Descriptions

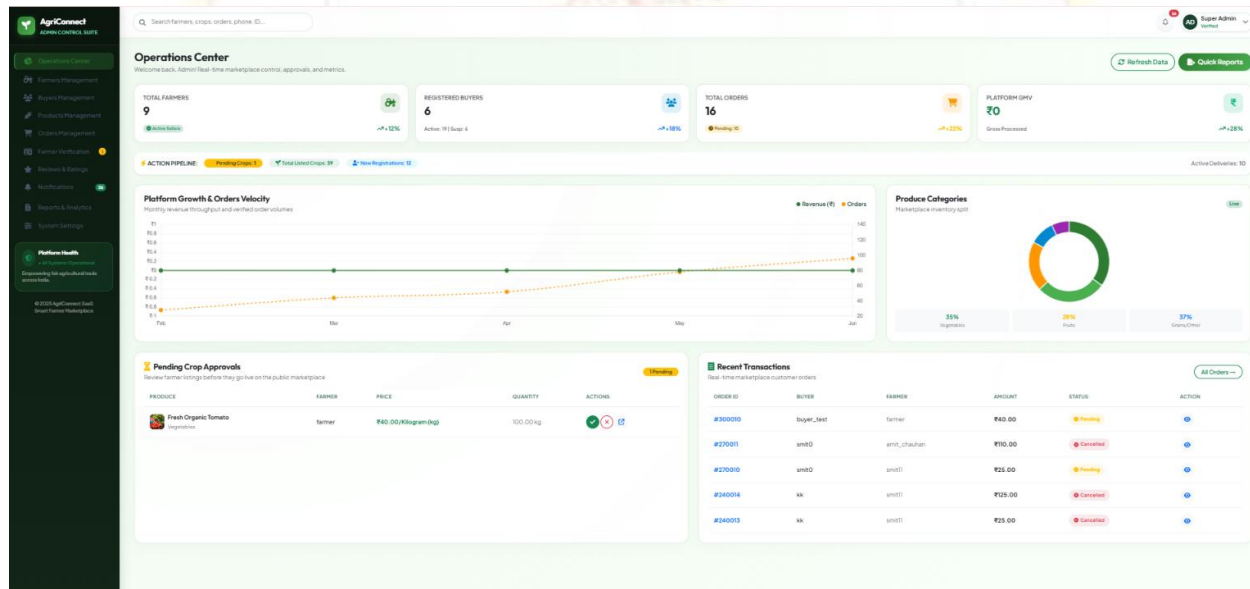
This section presents visual screen captures and detailed functional descriptions for all major modules of the AgriConnect system across Farmer, Buyer, and Admin portals.

9.1 Login & Authentication Screen



Multi-role secure authentication portal supporting login for Farmers, Buyers, and System Administrators with input validation and session tracking.

9.2 Admin Control Dashboard



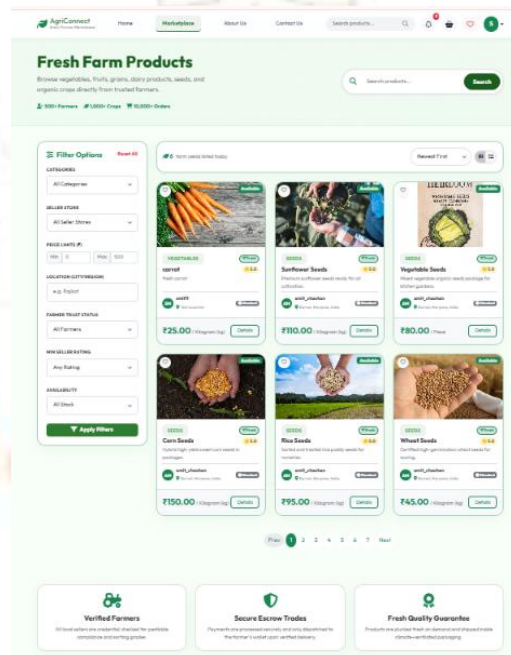
Centralized administrative management dashboard displaying platform user metrics, crop approval queues, revenue summary, and system report generation.

9.3 Farmer Management Dashboard

#	FARMER NAME	CONTACT	LOCATION	CROP LISTED	STATUS	ACCOUNT ACTIONS
1	farmer Green Valley Organic Farm	farmer@agricconnect.org	-	10kgs	Verified	
2	ramesh_patel Green Valley Farms	ramesh@agricconnect.org	-	50kgs	Pending	
3	suresh_kumar Fresh Harvest Farm	suresh@agricconnect.org	-	30kgs	Pending	
4	mahesh_singh Organic Earth Farm	mahesh@agricconnect.org	-	40kgs	Pending	
5	rajesh_verma Sustainable Agriculture	rajesh@agricconnect.org	-	20kgs	Pending	
6	amit_chaudhan Green Leaf Farms	amit@agricconnect.org	-	15kgs	Pending	
7	amit amit's Farm	amit@agricconnect.org	-	10kgs	Verified	
8	testfarmer123 Test Farm	testfarmer123@agricconnect.org	-	20kgs	Pending	
9	amit1 Test Farm	amit1@agricconnect.org	Test City	10kgs	Pending	

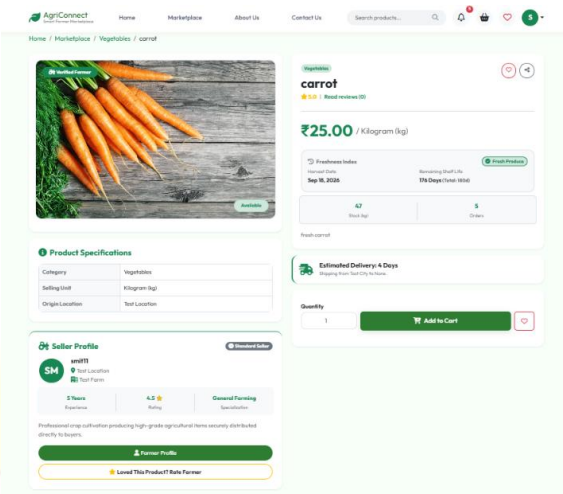
Farmer management portal providing harvest stock oversight, incoming buyer orders tracking, price updating tools, and revenue analytics.

9.4 Buyer Marketplace & Crop Catalog



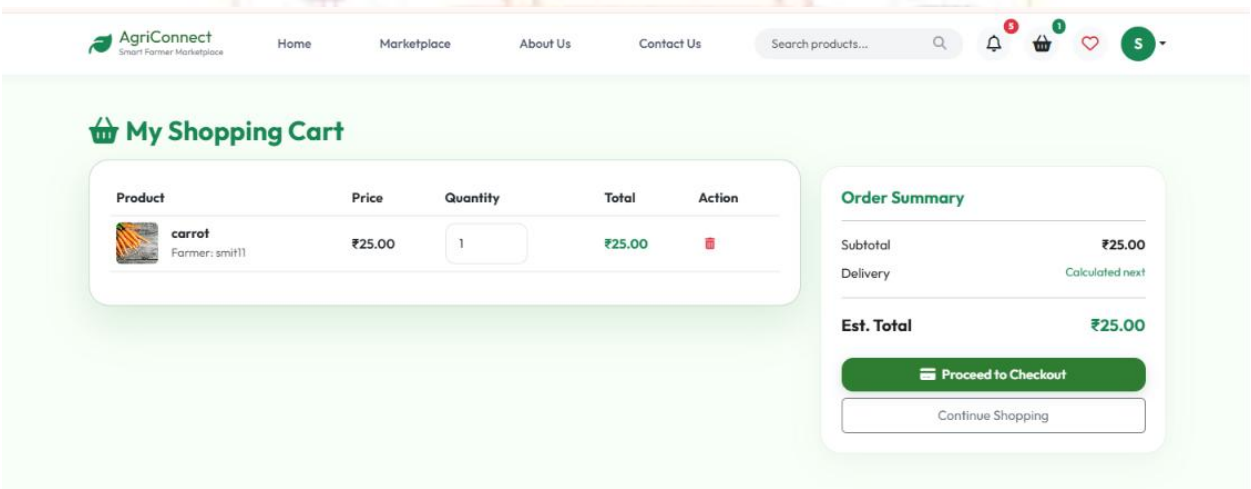
Dynamic produce catalog presenting available crops with high-resolution imagery, category filtering, search, price per kg, and stock availability.

9.5 Crop Detail & Quantity Selector



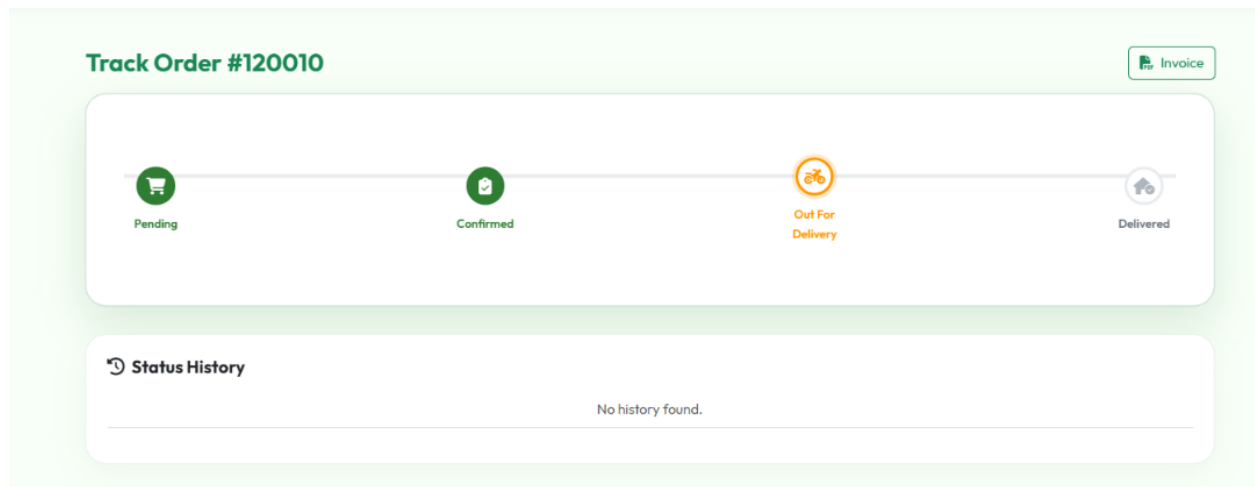
Detailed crop view displaying harvest date, shelf-life indicators, farmer verification badge, dynamic quantity calculator, and Add-to-Cart controls.

9.6 Shopping Cart & Checkout Page



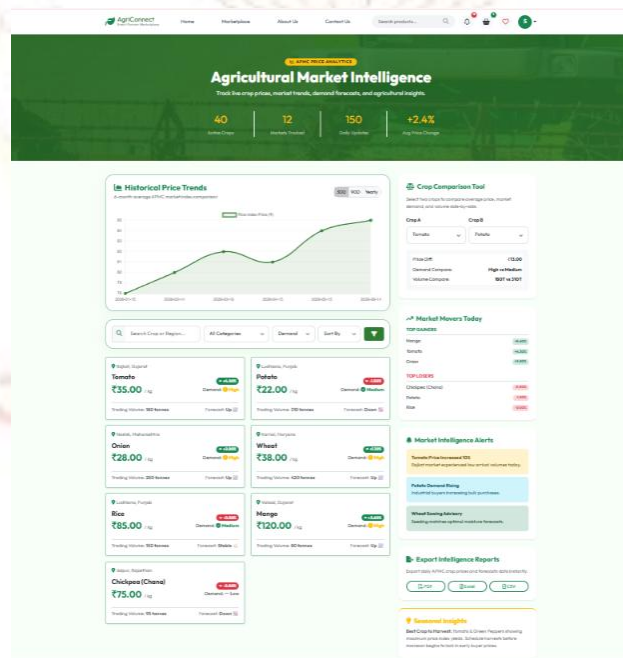
Shopping cart summary with itemized pricing, delivery address selection, multi-mode payment options (COD/UPI), and OTP verification step.

9.7 Order Tracking & History Page



Order tracking dashboard allowing buyers and farmers to monitor real-time shipment status, download PDF tax invoices, and submit return requests.

9.8 Market Price Trends & Analytics



Analytical insight dashboard providing historical crop price trends, market demand forecasts, and price fluctuation statistics to guide farmers.

9.9 order confirm otp verification

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Verify Your Order

Secure Checkout Processing

We have sent a 6-digit verification code to:

smitpithadiya29@gmail.com

Valid for 04:54

 **Verify & Place Order**

Didn't receive the code?

 **Resend OTP**

After the order is confirmed, an OTP verification message will be sent to your email and the order will be completed.

10. Test Cases

Comprehensive software testing was conducted across all system modules to verify user input validation, data integrity, security controls, and functional correctness. All test cases yielded expected outputs and passed successfully.

Test Suite 1: Login & User Authentication

Field / Module	Test Case Description	Expected Output	Actual Output	Result
Username & Password	Display input fields for username & password	System renders two clean input text fields	Fields rendered cleanly	Pass
Login Button	Verify login with valid credentials	User authenticated & redirected to role dashboard	Redirected to dashboard	Pass
Validation	Check empty field submission	Displays error: 'Username/Password required'	Error displayed	Pass
Invalid Credentials	Check login with incorrect password	Displays error: 'Invalid username or password'	Error displayed	Pass

Test Suite 2: Farmer Crop Management

Field / Module	Test Case Description	Expected Output	Actual Output	Result
Add Crop	Test adding new crop with price & stock	New crop added and set to pending approval	Crop added successfully	Pass
Edit Crop	Update crop price per kg and stock quantity	Crop listing updated in database and marketplace	Listing updated	Pass
Stock Status	Set crop quantity to 0	Listing automatically marked 'Out of	Marked out of stock	Pass

	Stock'			
Delete Crop	Remove crop listing from farmer portal	Crop entry removed from catalog	Removed successfully	Pass

Test Suite 3: Shopping Cart & Checkout

Field / Module	Test Case Description	Expected Output	Actual Output	Result
Add to Cart	Select crop quantity and click Add to Cart	Crop added to user cart with total calculated	Added to cart	Pass
Update Quantity	Increase item quantity in shopping cart	Subtotal and grand total dynamically update	Totals updated	Pass
Remove Item	Click delete icon on cart item	Item removed from cart	Item removed	Pass
Checkout OTP	Submit order with phone number	System generates 6-digit OTP for order confirmation	OTP sent & verified	Pass

Test Suite 4: Admin Control & Approval

Field / Module	Test Case Description	Expected Output	Actual Output	Result
Crop Approval	Admin approves pending farmer crop listing	Crop status changed to Approved & visible in marketplace	Status updated to Approved	Pass
User Role Audit	Admin inspects registered users list	Displays all active farmers, buyers, and admin accounts	List displayed cleanly	Pass

System Reports	Click Generate Sales Report PDF	PDF invoice/report generated and downloaded	Report generated	Pass
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11. Limitations

- **Internet Connectivity Dependency:** The application requires stable internet access for real-time market price updates and order placements, which may pose challenges in remote rural regions with limited network coverage.
- **Manual Inventory Updates:** Farmers must manually update crop availability and harvested quantities, leading to potential discrepancies if stock updates are delayed after local offline sales.
- **Scalability for Enterprise Volumes:** While the current Django architecture easily handles thousands of concurrent users, massive enterprise scaling across multiple geographical zones requires additional cloud load balancing and redis caching.

12. Future Enhancements

- **AI-Driven Crop Disease Identification:** Integrate machine learning models allowing farmers to upload leaf photos to automatically diagnose crop diseases and receive treatment remedies.
- **Multilingual Regional Voice Support:** Introduce voice-assisted navigation in Hindi, Gujarati, Marathi, and regional languages to assist non-literate farmers.
- **Weather & Price Predictive Analytics:** Incorporate real-time weather API feeds and AI price prediction forecasting to help farmers choose optimal harvest times.
- **Native Mobile Application:** Expand AgriConnect to native Android and iOS mobile applications using Flutter for seamless mobile notifications and offline capability.

13. Bibliography

- Django Web Framework Documentation: <https://docs.djangoproject.com/>
- Python Official Documentation: <https://www.python.org/>
- Bootstrap 5 Documentation: <https://getbootstrap.com/>
- Stack Overflow Technical Q&A: <https://stackoverflow.com/>
- Draw.io Diagramming Tool: <https://app.diagrams.net/>
- MDN Web Docs (HTML/CSS/JS): <https://developer.mozilla.org/>